

84. Stem Cells, Kidney Regeneration, and Gene and Cell Therapy in Nephrology · assets 1-2 / 2

Table 84.1 Classification of Stem Cells	
Stem Cells	Characterizations and Functions
Embryonic stem cells (ESCs)	Human ESCs can differentiate into any cell type (pluripotent) and potentially be used for various medical conditions including tissue injury. However, application is limited due to ethical reservations.
Adult stem cells (ASCs)	ASCs, also known as somatic stem cells, are undifferentiated cells existing in various tissues and can differentiate into different cell types. ASCs include hematopoietic stem cells and mesenchymal stem cells (MSCs).
Induced pluripotent stem cells (iPSCs)	iPSCs are generated by genetic reprogramming through introducing specific genes into adult cells to acquire characteristics of embryonic stem cells. iPSCs generated from patient's own cells do not cause immunologic reactions when used as autotherapy. Thus iPSCs have potential in personalized medicine and tissue repair.
Progenitor cells	These cells exhibit differentiation potential between stem cells and mature cells. These cells play a part in maintaining the wound healing process to replace the damaged or dead cells, participating in the normal cell turnover of organs.

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Table 84.2 Currently Ongoing Clinical Trials of Stem Cell Therapy in Chronic Kidney Disease				
Trial ID	Countries	Title	Link	Location
NCT01125229	USA	Unilateral Cord Mesenchymal Stem Cell Transplantation for Treating Single Kidney end-stage renal disease	https://clinicaltrials.gov/ct2/show/study/NCT01125229	Kowloon, China
NCT01761624	USA	Human Mesenchymal Stem Cell Transplantation to Treat End-stage Renal Disease	https://clinicaltrials.gov/ct2/show/study/NCT01761624	Zurich, Switzerland
NCT04886701	USA	Stem Cell Therapy for Chronic Kidney Disease	https://clinicaltrials.gov/ct2/show/study/NCT04886701	Richmond, Virginia, USA
NCT03513988	Japan	Human Umbilical Stem Cell (UC-MSC) Derived Cell	https://clinicaltrials.gov/ct2/show/study/NCT03513988	Wuhan, China
NCT01981840	USA	Chronic Kidney Disease (CKD) Stem Cell Transplantation	https://clinicaltrials.gov/ct2/show/study/NCT01981840	St. John's, Antigua and Barbuda
NCT02032041	USA	Stem Cell Therapy for CKD	https://clinicaltrials.gov/ct2/show/study/NCT02032041	Dallas, Dergason
NCT04382036	USA	APOL1 or Not? Apolipoprotein B-48	https://clinicaltrials.gov/ct2/show/study/NCT04382036	Jacksonville, Florida, USA
NCT03873148	Spain	Human Umbilical Stem Cell Transplantation	https://clinicaltrials.gov/ct2/show/study/NCT03873148	Valencia, Spain
NCT03080809	USA	Human Umbilical Stem Cell Transplantation for Chronic Kidney Disease	https://clinicaltrials.gov/ct2/show/study/NCT03080809	Palo Alto, California, USA
NCT04011717	China	The Study of Effects and Safety of Human Umbilical Cord MSCs in End-stage Renal Disease	https://clinicaltrials.gov/ct2/show/study/NCT04011717	Nanjing, China
NCT03080878	Spain	Human Umbilical Stem Cell Transplantation for Chronic Kidney Disease	https://clinicaltrials.gov/ct2/show/study/NCT03080878	Nanjing, China

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